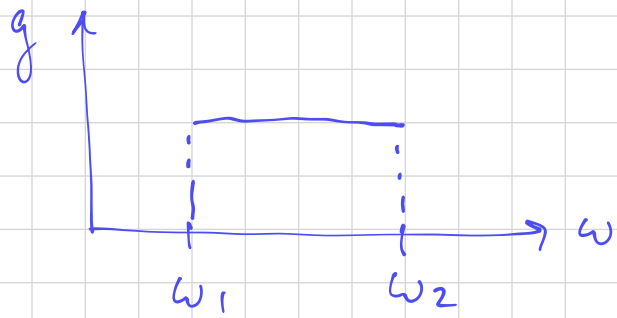
Prob. 1 $\quad g(\omega) = A N \, \theta(\omega_2 - \omega)\, \theta(\omega - \omega_1)$



$$E = \int_0^\infty d\omega \; g(\omega)\, \hbar\omega \left(n_B(\omega) + \frac{1}{2} \right)$$

$$n_B(\omega) = \frac{1}{e^{\beta\hbar\omega} - 1}$$

(a) High T means $\beta\hbar\omega = \dfrac{\hbar\omega}{k_B T} \ll 1$ for all ω.

This means $\hbar\omega_2 \ll k_B T$.

In this limit we can expand $n_B \approx \dfrac{1}{1 + \beta\hbar\omega - 1} = \dfrac{k_B T}{\hbar\omega}$

$$\Rightarrow E = \int_{\omega_1}^{\omega_2} d\omega \; A N \, \hbar\omega \left(\frac{k_B T}{\hbar\omega} + \frac{1}{2} \right) \simeq A N k_B T \int_{\omega_1}^{\omega_2} d\omega$$

$$= A N k_B T \, (\omega_2 - \omega_1)$$

On the other hand, $E = 3 N k_B T$

$$\Rightarrow 3 N k_B T = A N k_B T \, (\omega_2 - \omega_1) \Rightarrow A = \frac{3}{\omega_2 - \omega_1}$$

$$C = \frac{\partial E}{\partial T} = 3 k_B \text{ per atom}$$

(b) Low T means $\beta\hbar\omega = \dfrac{\hbar\omega}{k_B T} \gg 1$ for all $\omega \Rightarrow k_B T \ll \hbar\omega_1$

In this limit $n_B(\omega) \approx e^{-\beta\hbar\omega}$

$$E(T) = A N \int_{\omega_1}^{\omega_2} d\omega \; \hbar\omega \, e^{-\beta\hbar\omega} = A N \hbar (\beta\hbar)^{-2} \int_{x_1}^{x_2} dx \; x \, e^{-x}$$

$$\beta\hbar\omega = x \qquad x_{1,2} = \beta\hbar\omega_{1,2}$$

$$E(T) = A N \frac{(k_B T)^2}{\hbar} \underbrace{\int_{x_1}^{x_2} dx \; x \, e^{-x}}_{-(1+x)e^{-x}\Big|_{x_1}^{x_2} \approx x_1 e^{-x_1} \text{ because } x_2 \gg x_1.}$$

$$E(T) \cong A N \frac{(k_B T)^2}{\hbar} \frac{\hbar\omega_1}{k_B T} e^{-\frac{\hbar\omega_1}{k_B T}} = N \frac{3\omega_1}{\omega_2-\omega_1} k_B T e^{-\frac{\hbar\omega_1}{k_B T}}$$

$$C(T) = \frac{\partial E}{\partial T} = \frac{3\omega_1}{\omega_2-\omega_1}\left(k_B e^{-\frac{\hbar\omega}{k_B T}} + k_B T \frac{\hbar\omega_1}{k_B T^2} e^{-\frac{\hbar\omega_1}{k_B T}}\right)$$

$$\approx k_B \frac{3\omega_1}{\omega_2-\omega_1} \frac{\hbar\omega_1}{k_B T} e^{-\frac{\hbar\omega_1}{k_B T}}$$ is exponentially suppressed.

(c) Intermediate T: $\hbar\omega_1 \ll k_B T \ll \hbar\omega_2$

$$E(T) = A N \int_{\omega_1}^{\omega_2} d\omega\, \hbar\omega \frac{1}{e^{\beta\hbar\omega}-1} = \hbar A N \left(\frac{k_B T}{\hbar}\right)^2 \int_{x_1}^{x_2} dx \frac{x}{e^x-1}$$

where $x_1 = \frac{\hbar\omega_1}{k_B T} \ll 1$ and $x_2 = \frac{\hbar\omega_2}{k_B T} \gg 1$.

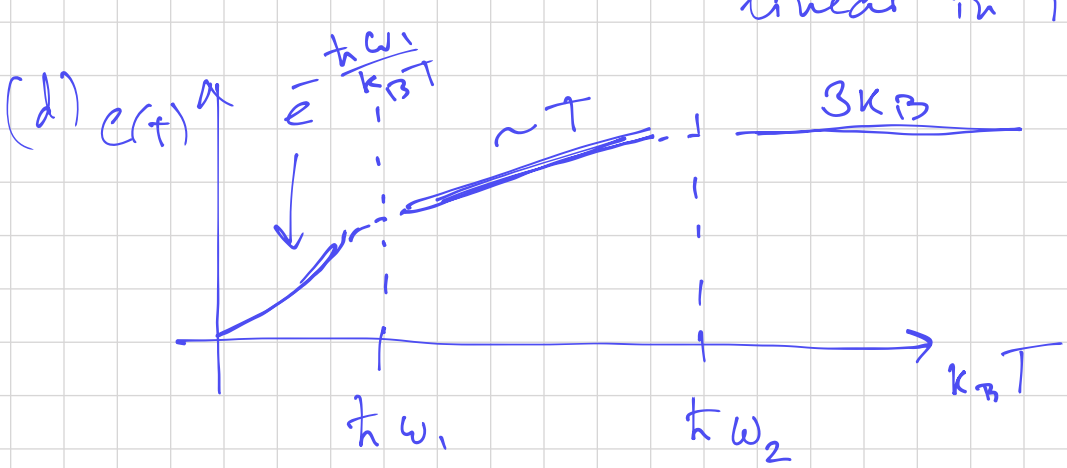
Therefore here $E(T) \simeq \frac{3N}{\omega_2-\omega_1} \frac{(k_B T)^2}{\hbar} \cdot \underbrace{\int_0^\infty \frac{dx\, x}{e^x-1}}_{J \text{ (number)}}$

$$\Rightarrow C(T) \approx 6 k_B \frac{k_B T}{\hbar(\omega_2-\omega_1)} \cdot J$$

per atom

linear in T!

(d)



Prob. 2 Transverse EM wave

$$-\nabla^2 \vec{E}(\omega, z) = \frac{\omega^2}{c^2} \varepsilon(\omega) \vec{E}(\omega, z)$$

$$\vec{E}(\omega, z) = \vec{E}(\omega) e^{ikz} \Rightarrow k^2 \vec{E}(\omega) e^{ikz} = \frac{\omega^2}{c^2} \varepsilon(\omega) \vec{E} e^{ikz}$$

$$\Rightarrow k^2 = \frac{\omega^2}{c^2} \varepsilon(\omega)$$

$$\varepsilon(\omega) = 1 + \frac{i\sigma(\omega)}{\varepsilon_0 \omega}, \quad \sigma(\omega) = \frac{\sigma_0}{1 - i\omega\tau}, \quad \sigma_0 = \frac{ne^2\tau}{m}$$

(a) $$\varepsilon(\omega) = 1 + \frac{i\sigma_0}{\varepsilon_0 \omega (1 - i\omega\tau)} = 1 + \frac{i\sigma_0}{\varepsilon_0 \omega (-i\omega\tau)\left(1 - \frac{1}{i\omega\tau}\right)}$$

Expand in $\omega\tau \gg 1$

$$\simeq 1 + \frac{i\sigma_0}{\varepsilon_0 \omega (-i\omega\tau)} \left(1 + \frac{1}{i\omega\tau} + \dots\right) = 1 - \frac{\sigma_0}{\varepsilon_0 \omega^2 \tau}$$

$$+ \frac{i\sigma_0}{\varepsilon_0 \omega (\omega\tau)^2} = 1 - \frac{ne^2}{\varepsilon_0 m} \frac{1}{\omega^2} + i \frac{ne^2}{\varepsilon_0 m} \frac{1}{\omega^2(\omega\tau)}$$

$$= \underbrace{1 - \frac{\omega_p^2}{\omega^2}}_{\varepsilon_1} + i \underbrace{\frac{\omega_p^2}{\omega^2(\omega\tau)}}_{\varepsilon_2}, \quad \text{where } \omega_p^2 = \frac{ne^2}{\varepsilon_0 m}.$$

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\,\varepsilon_2(\omega).$$

(b) At $\omega = \omega_p$: $\varepsilon_1(\omega_p) = 1 - 1 = 0$, $\varepsilon_2(\omega_p) = \frac{1}{\omega_p \tau} \ll 1$ but finite.

(c) $$k = k_1 + i k_2 \Rightarrow (k_1 + i k_2)^2 = \frac{\omega_p^2}{c^2} \varepsilon(\omega_p) =$$

$$= \frac{\omega_p^2}{c^2} \left(\varepsilon_1(\omega_p) + i\,\varepsilon_2(\omega_p)\right)$$

$$k_1^2 - k_2^2 + i\,2k_1 k_2 = \frac{\omega_p^2}{c^2} \frac{i}{\omega_p \tau} = i \frac{\omega_p}{\tau c^2}.$$

Right-hand side is proportional to i

$$\Rightarrow \quad k_1^2 - k_2^2 = 0 \quad \Rightarrow \quad k_1 = k_2 .$$

$$2 k_1 k_2 = 2 k_1^2 = \frac{\omega_p}{\tau c^2} \quad \Rightarrow \quad k_1 = \sqrt{\frac{\omega_p}{2 c^2 \tau}} = k_2 .$$

Therefore $\vec{E}(\omega_p, z) = \vec{E}(\omega)\, e^{i(k_1 + i k_2) z} =$

$$= \vec{E}(\omega)\, e^{i k_1 z}\, e^{-k_2 z}$$

It decays exponentially on distance $\xi = \frac{1}{k_2} = \sqrt{\frac{2 c^2 \tau}{\omega_p}}$

$$= c\tau \sqrt{\frac{2}{\omega_p \tau}} .$$